

A2: Evaluation Plan for Conversations Gone Awry - Change My View Dataset

Franklin Li

Corpus: CGA - CMV (Small) Dataset, where each conversation ending in moderate action is matched with a same-topic counterpart that did not veer off-course (Chang & Danescu-Niculescu-Mizil, 2019); extracted from a forum (Change My View subreddit) where users post opinions on often-contentious topics and actively solicit counterarguments with the goal of reconsidering or shifting their views (Reddit, 2025); dataset available through ConvoKit version 3.5.0 (Chang et al., 2020, 2025)

a) Best K via coherence (LDA)

- K grid attempted: range (4, 45, 5) → range (16, 25, 1)
- Coherence metric and settings: NPMI, window_size = 20, top-n = 10, random_state = 42
- Library/API + version: Python 3.12.5, Gensim 4.4.0 (LdaModel, CoherenceModel, n-gram parsing where $1 \leq n \leq 3$; Řehůřek & Sojka, 2010), spaCy 3.8.7 (lemmatization, aggressive stopword filtering), pandas 2.3.3 (data wrangling), contractions 0.1.73 (text normalization; van Kooten, 2021)

Justification of Chosen K: Coherence peaked at K=20 (NPMI = 0.0711) when number of topics ranged from 5 to 40 (increments of 5). The same apex remained after the grid was narrowed down to a range of 16 to 25 (increments of 1).

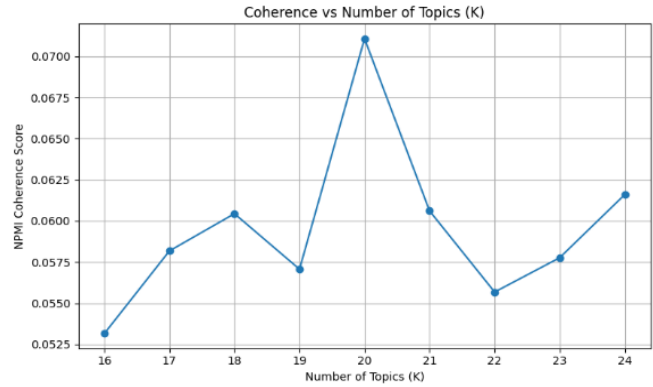


Figure 1. Coherence drastically drops when topic quantity deviates even slightly from K = 20.

b) Double-check at K = 20

- Runs: Seeds = [562, 324, 136, 571, 729], preprocessing same as above, except random_state = seed
- Topic overlap method: Intra-run (Seed = 324) Jaccard similarity on top-10 terms = 0.015, avg. across all 5 seeds = 0.017 ± 0.0023
- Cross-run stability: Hungarian matching w/ Jaccard similarity on top-10 terms, avg. alignment = 0.32 ± 0.037
- Library/API + version: Python 3.12.5, Gensim 4.4.0 (Řehůřek & Sojka, 2010), SciPy 1.16.2 (Hungarian algorithm), seaborn 0.13.2 (heatmap visualization)

Stability summary: Topics moderately aligned across the five runs (32% Jaccard similarity across top-10 terms). The best-performing run (seed = 324; NPMI = 0.041) yielded minimal inter-topic redundancy (maximum duplication rate of 0/20 topics, Jaccard ≥ 0.30) and low intra-run topic overlap (1.5% Jaccard similarity across top-10 terms). Although stability is on the lower end, K = 20 is retained on the assumption that online discursive contexts may produce high noise, which can plausibly derail cross-run alignment.

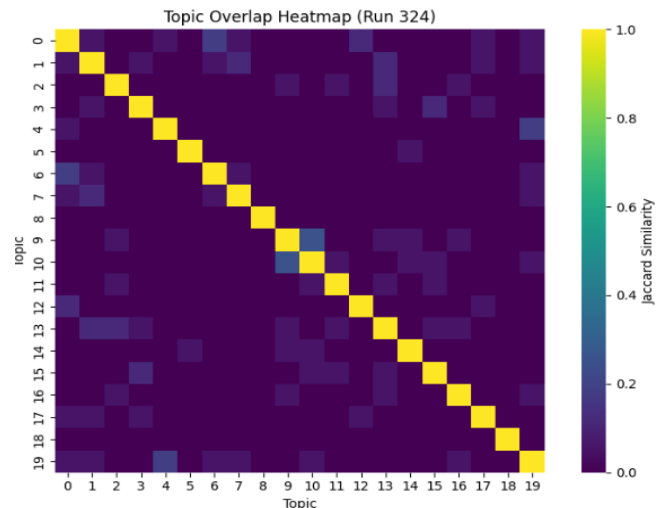


Figure 2. Elevated similarity between topics 9 and 10 (diagonal blue clusters at centre) warrants closer inspection. For the most part, the abundance of purple clusters signifies low intra-run topic overlap.

c) Reflexive Thematic Analysis at K = 20

- Topics selected = 3 (domain-salient), 7 (clear-cut), 9 (ambiguous)
- Materials: words per topic = 15; docs per topic = 12; borderline docs = 3
- Procedure: 1) Lived positionality & implicit assumptions were laid out prior to familiarization with documents with 12 highest topic probabilities (Braun & Clarke, 2006), 2) Initial codes were constructed in close consultation with supporting and refuting evidence and underwent iterative refinement following careful re-readings before culminating in the proposition of constituent codes, 3) based on these propositions, themes were constructed prior to their reassessment, characterization, and labelling (Braun & Clarke, 2006)
- Audit Trail: Top documents for all topics and borderline documents for topic 9 were annotated with reflexive memos as initial codes emerged, metamorphosed, and informed the construction of categories, which influenced the identification, formation, and characterization of themes.

Reflexivity Note: As a citizen of a country where the right to bear arms is heavily restricted, I acknowledge the biases that I hold against the possession of weapons of deadly force, which I counteract by exercising controlled detachment and emboldened awareness of differing political stances on relevant legislative frameworks. As a recent employee at UofT's Centre for Global Health, I engaged in work grounded in values aligned with reproductive rights advocacy; at the same time, I refrain from immediately invalidating opposing viewpoints while remaining mindfully dispassionate and cognizant of my positionality. As a currently unemployed and moderately socioeconomically privileged graduate student, I approach this investigation with a transparent naivety which motivates me to uncover discursive insights into the current & future state of the global economy.

T3 (domain-salient): gun, kill, shoot, weapon, law, firearm, ban, police, state, murder, gun_control, time, death, crime, criminal; Candidate Theme: Legalization Versus Criminalization of Arms Possession
Debate around prescriptive measures to restrict arms possession broaches cost-benefit analyses of gun ownership in light of lethality rates (accidental vs. intentional; homicidal vs. injurious), geographical differences in legislation, and justifications for self-defence.

T7 (clear-cut): child, kid, parent, life, abortion, choice, person, family, risk, baby, adult, sex, decision, mother, time; Candidate Theme: Reproductive Autonomy Versus Life Preservation
Back-and-forth dialogue unambiguously centres references to adoption as an alternative, decisions leading up to pregnancies, child support, and a maternal-paternal differential in the amount of influence that each parent exerts over the pregnancy outcome.

T9 (ambiguous): pay, work, job, company, time, buy, employee, cost, business, tip, place, money, employer, live, sell; Candidate Theme: Logistical Factors Influencing Financial Decisions
Top words shared with T10 include "money", "pay", "job", "business", and "work" - terms pertinent to organizational norms and economic transactions. Top documents surface contentious debate over tipping culture. Borderline documents surface sentiments around illicit immigration, automotive ownership, and gender disparities in the division of labour.

Interpretive Judgement: Lexical evidence for T3 exemplifies high interpretability and pertinence to the domain in question - polemical Internet discourse potentially resulting in moderative action.
Conversations about T7 foreground layers of nuance that economic factors and parenting capabilities lend to justifications for or against abortive decision-making. The relationship between T9 and borderline documents suggests that generalized constructs (e.g., labour investment and economic logistics) are somewhat diffuse, yet moderately pertinent to other topics extracted from the corpus.

Footnote (tooling disclosure): ConvoKit 3.5.0 (Chang et al., 2020, 2025), Python 3.12.5 with pandas 2.3.3, NumPy 2.3.4, SciPy 1.16.2, matplotlib 3.10.7, seaborn 0.13.2, spaCy 3.8.7, contractions 0.1.73 (van Kooten, 2021), and Gensim 4.4.0 (Řehůřek & Sojka, 2010; LDA with passes = 10, alpha = ‘symmetric’, eta = ‘symmetric’); seeds = [562, 324, 136, 571, 729]. Coherence = NPMI (window = 20). Topic overlap method: Jaccard on top-10 terms. Cross-run alignment check: Hungarian algorithm.

ConvoKit is funded in part by the U.S. National Science Foundation under Grant No. IIS-1750615 (CAREER). Any opinions, findings, and conclusions in this work are those of the author(s) and do not necessarily reflect the views of Cornell University or the National Science Foundation.

References

- Braun, V., & Clarke, V. (2006, January). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77–101. <https://doi.org/10.1191/1478088706qp063oa>
- Chang, J.P., Chiam, C., Fu, L., Wang, A.Z., Zhang, J., & Danescu-Niculescu-Mizil, C. (2020, July). ConvoKit: A toolkit for the analysis of conversations. *Proceedings of the 21th Annual Meeting of the Special Interest Group on Discourse and Dialogue*, 57-60. <https://doi.org/10.18653/v1/2020.sigdial-1.8>
- Chang, J.P., Chiam, C., Fu, L., Wang, A.Z., Zhang, J., & Danescu-Niculescu-Mizil, C. (2025, October 15). *ConvoKit* (Version 3.5.0) [Software library]. <https://convokit.cornell.edu/>
- Chang, J.P., & Danescu-Niculescu-Mizil, C. (2019, November). Trouble on the horizon: Forecasting the derailment of online conversations as they develop. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, 4743-4754. <https://doi.org/10.18653/v1/d19-1481>
- Reddit. (2025, November 13). *r/changemyview*. <https://www.reddit.com/r/changemyview/>
- Řehůřek, R., & Sojka, P. (2010, May 22). Software framework for topic modelling with large corpora. *Proceedings of the LREC 2010 Workshop on New Challenges for NLP Frameworks*, 46-50. <https://doi.org/10.13140/2.1.2393.1847>
- van Kooten, P. (2021). *Contractions* (Version 0.1.73) [Python package]. GitHub. <https://github.com/kootenpv/contractions>